1. Jelaskan kelebihan dan kekurangan dari routing protocol OSPF!
   a) Kelebihan
      - Konvergensi Cepat: Menggunakan Algoritma Dijkstra (Shortest Path First - SPF) untuk menghitung jalur tercepat.
      - Skalabilitas Tinggi: Mendukung pembagian jaringan menjadi Area (multi-area) yang mengurangi beban routing.
      - Jalur Optimal: Selalu memilih jalur terpendek dan paling efisien berdasarkan cost (biaya).
      - Load Balancing: Mendukung equal-cost load balancing secara default.
   b) Kekurangan
      - Kompleksitas Konfigurasi: Lebih rumit dibandingkan RIP, membutuhkan pemahaman tentang Area, LSA, dan jenis router.
      - Membutuhkan Sumber Daya Tinggi: Memerlukan lebih banyak Memori (untuk menyimpan Link-State Database) dan CPU (untuk perhitungan SPF).
      - Perencanaan Ketat: Membutuhkan desain jaringan yang cermat, terutama dalam penentuan Area 0 (Backbone).

2. Gunakan Protocol OSPF untuk menghubungkan semua network!

Router 0:

```
Router>enable
Router#conf t
Router(config)#hostname R0
R0(config)#int gig0/0/0
R0(config-if)#ip add 192.168.10.1 255.255.255.0
R0(config-if)#no shutdown
R0(config-if)#int se0/1/0
R0(config-if)#ip add 10.10.10.1 255.255.255.128
R0(config-if)#no shutdown
```

```
R0(config-if)#ex
R0(config)#router ospf 10
R0(config-router)#network 192.168.10.0 0.0.0.255 area 0
R0(config-router)#network 10.10.10.0 0.0.0.127 area 0
```

Router 1:

```
Router#enable
Router#conf t
Router(config)#hostname R1
R1(config)#int gig0/0/0
R1(config-if)#ip add 192.168.20.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#int se0/1/0
R1(config-if)#ip add 10.10.10.2 255.255.255.128
R1(config-if)#no shutdown
R1(config-if)#int se0/1/1
R1(config-if)#ip add 20.20.20.1 255.255.255.128
R1(config-if)#no shutdown
R1(config-if)#ex
R1(config)#route ospf 10
R1(config-router)#network 192.168.20.0 0.0.0.255 area 0
R1(config-router)#network 20.20.20.0 0.0.0.127 area 0
R1(config-router)#network 10.10.10.0 0.0.0.127 area 0
```

Router 2:

```
Router>enable
Router#conf t
Router(config)#hostname R2
R2(config)#int se0/1/1
R2(config-if)#ip add 20.20.20.2 255.255.255.128
R2(config-if)#no shutdown
```

```
R2(config-if)#int  se0/1/0
R2(config-if)#ip  add  30.30.30.1  255.255.255.128
R2(config-if)#no  shutdown
R2(config-if)#ex
R2(config)#route  ospf  10
R2(config-router)#network  30.30.30.0  0.0.0.127  area  0
R2(config-router)#network  20.20.20.0  0.0.0.127  area  0
```

Router 3:

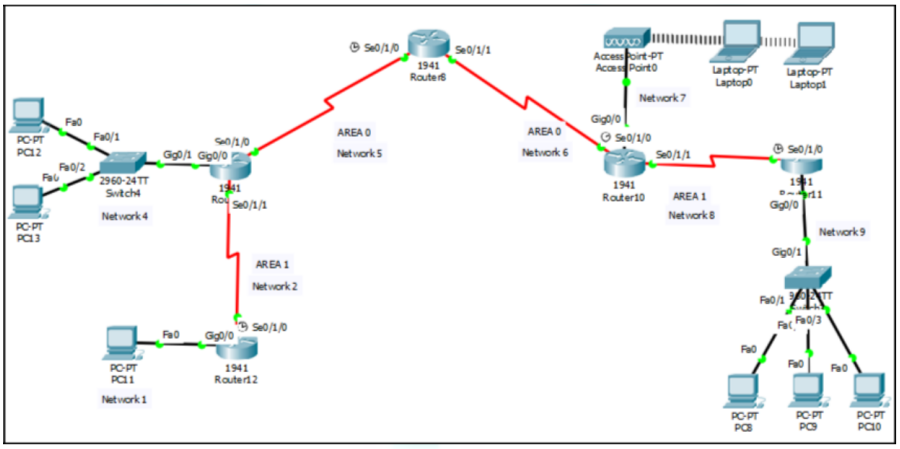
```
Router>enable
Router#conf  t
Router(config)#hostname  R3
R3(config)#int  gig0/0/0
R3(config-if)#ip  add  192.168.30.1  255.255.255.0
R3(config-if)#no  shutdown
R3(config-if)#int  se0/1/0
R3(config-if)#ip  add  30.30.30.2  255.255.255.128
R3(config-if)#no  shutdown
R3(config-if)#int  se0/1/1
R3(config-if)#ip  add  40.40.40.1  255.255.255.128
R3(config-if)#no  shutdown
R3(config-if)#ex
R3(config)#route  ospf  10
R3(config-router)#network  30.30.30.0  0.0.0.127  area  0
R3(config-router)#network  192.168.30.0  0.0.0.255  area  0
R3(config-router)#network  40.40.40.0  0.0.0.127  area  0
```

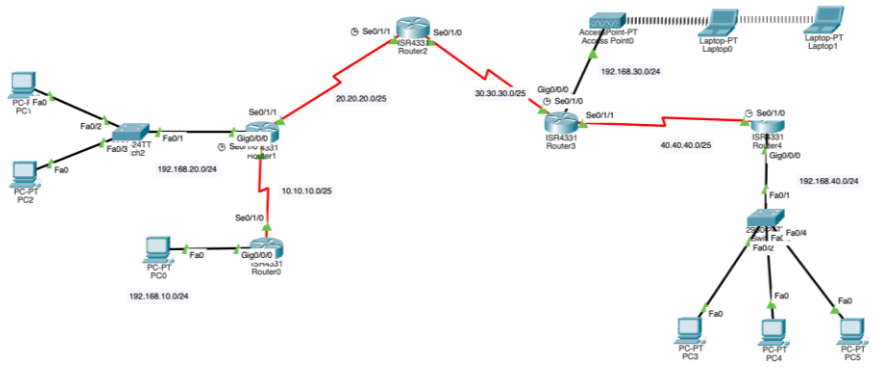
Router 4:

```
Router>enable
Router#conf  t
Router(config)#hostname  R4
```

```
R4(config)#int  gig0/0/0
R4(config-if)#ip  add  192.168.40.1  255.255.255.0
R4(config-if)#no  shutdown
R4(config-if)#int  se0/1/0
R4(config-if)#ip  add  40.40.40.2  255.255.255.128
R4(config-if)#no  shutdown
R4(config-if)#ex
R4(config)#route  ospf  10
R4(config-router)#network  192.168.40.0  0.0.0.255  area  0
R4(config-router)#network  40.40.40.0  0.0.0.127  area  0
```

3. Aktifkan DHCP Server agar semua Client bisa mendapatkan IP dari Router.





Router 0:

```
R0>enable
R0#conf t
R0(config)#ip dhcp pool R0
R0(dhcp-config)#network 192.168.10.0 255.255.255.0
R0(dhcp-config)#default 192.168.10.1
R0(dhcp-config)#dns-server 8.8.8.8
```

Router 1:

```
R1>enable
R1#conf t
R1(config)#ip dhcp pool R1
R1(dhcp-config)#network 192.168.20.0 255.255.255.0
R1(dhcp-config)#default 192.168.20.1
R1(dhcp-config)#dns-server 8.8.8.8
```

Router 3:

```
R3>enable
R3#conf t
R3(config)#ip dhcp pool R3
R3(dhcp-config)#network 192.168.30.0 255.255.255.0
R3(dhcp-config)#default 192.168.30.1
R3(dhcp-config)#dns-server 8.8.8.8
```

Router 4:

```
R4>enable
R4#conf t
R4(config)#ip dhcp pool R4
R4(dhcp-config)#network 192.168.40.0 255.255.255.0
R4(dhcp-config)#default 192.168.40.1
R4(dhcp-config)#dns-server 8.8.8.8
```

Ping Hasil Pengujian:

Ping PC2 ke PC0

C:\>ping 192.168.10.2

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=2ms TTL=126
Reply from 192.168.10.2: bytes=32 time=1ms TTL=126
Reply from 192.168.10.2: bytes=32 time=1ms TTL=126
Reply from 192.168.10.2: bytes=32 time=3ms TTL=126

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 3ms, Average = 1ms
```

Ping PC1 ke PC5

C:\>ping 192.168.40.3

```
C:\>ping 192.168.40.3

Pinging 192.168.40.3 with 32 bytes of data:

Reply from 192.168.40.3: bytes=32 time=32ms TTL=124
Reply from 192.168.40.3: bytes=32 time=4ms TTL=124
Reply from 192.168.40.3: bytes=32 time=36ms TTL=124
Reply from 192.168.40.3: bytes=32 time=44ms TTL=124

Ping statistics for 192.168.40.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 44ms, Average = 29ms
```

Ping PC0 ke Laptop1

C:\>ping 192.168.30.4

```
C:\>ping 192.168.30.4

Pinging 192.168.30.4 with 32 bytes of data:

Reply from 192.168.30.4: bytes=32 time=12ms TTL=124
Reply from 192.168.30.4: bytes=32 time=11ms TTL=124
Reply from 192.168.30.4: bytes=32 time=15ms TTL=124
Reply from 192.168.30.4: bytes=32 time=14ms TTL=124

Ping statistics for 192.168.30.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 15ms, Average = 13ms
```

Ping Laptop0 ke PC2

C:\>ping 192.168.20.2

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=44ms TTL=125
Reply from 192.168.20.2: bytes=32 time=32ms TTL=125
Reply from 192.168.20.2: bytes=32 time=52ms TTL=125
Reply from 192.168.20.2: bytes=32 time=28ms TTL=125

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 28ms, Maximum = 52ms, Average = 39ms
```

Ping PC4 ke PC1

C:\>ping 192.168.20.3

```
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Reply from 192.168.20.3: bytes=32 time=7ms TTL=124
Reply from 192.168.20.3: bytes=32 time=53ms TTL=124
Reply from 192.168.20.3: bytes=32 time=3ms TTL=124
Reply from 192.168.20.3: bytes=32 time=3ms TTL=124

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 53ms, Average = 16ms
```